Final Viva Presentation & Live Demo Defense Guide

Course: Report Writing (21CSC402P) / PBL Final Assessment

Date of Assessment: 10 November 2026 (20 Marks: Report & Viva with Demo)

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Project Title: Reliability-Aware Audio-Visual Fusion for Generalizable Deepfake Detection

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1. VIVA PRESENTATION STRUCTURE & TIMELINE (12–15 MINUTES)

To score the maximum 20 marks in the Final Viva and Demo, structure your oral presentation as follows:

15-MINUTE VIVA PRESENTATION SCHEDULE		
Time Window	**Topic / Focus**	**Key Message to Examiners**
00:00 – 02:00 min	**1. Problem Definition & Research Gap**	**Why unimodal fails & why naive**
		equal-weight fusion is brittle
02:00 – 04:30 min	**2. Proposed Architecture & Formulations**	**MobileNetV3 + DS-CNN + Quality**
		& Sync + Reliability MLP
04:30 – 07:00 min	**3. Data Integrity & Leakage Prevention**	**Zero identity overlap across**
		21,544 FakeAVCeleb clips
07:00 – 10:00 min	**4. LIVE SOFTWARE DEMO (Interactive)**	**Run local web UI, upload clip,**
		explain (p_v, p_a, w_v, w_a)
10:00 – 12:30 min	**5. Experimental Analysis & Findings**	**Explain why equal fusion leads**
		in Epoch 1 & why MLAAD drops

12:30 – 15:00 min	6. Conclusion, Limitations & Viva Q&A	Honest scientific defense; no
		unmeasured SOTA claims

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2. STEP-BY-STEP LIVE DEMO EXECUTION GUIDE

The project includes an in-house browser demonstration system (`scripts/serve_web.py` + `web/index.html`). Follow this exact sequence during the live presentation.

Step 1: Pre-Viva Environment Readiness Check

Open a terminal in the project directory before entering the examination hall:

```
# 1. Activate project virtual environment
source .venv/bin/activate

# 2. Verify all 95 CPU unit tests pass
make test

# 3. Verify checkpoint integrity
ls -lh artifacts/checkpoints/reliability_fusion/best_model.pt
```

Step 2: Launch the Demonstration Server

Execute the CLI server:

```
python scripts/serve_web.py --checkpoint
artifacts/checkpoints/reliability_fusion/best_model.pt
```

Expected Server Output:

```
2026-10-27 00:04:12,105 | serve_web | INFO | Serving review interface at
http://127.0.0.1:7860
2026-10-27 00:04:12,106 | serve_web | INFO | Loaded checkpoint:
artifacts/checkpoints/reliability_fusion/best_model.pt
2026-10-27 00:04:12,106 | serve_web | INFO | Validation threshold: 0.2950 (fitted on
FakeAVCeleb validation)
```

Step 3: Open the Web Application

Open Chromium or any modern browser at:

```
http://127.0.0.1:7860
```

Point out the following key UI components to the examiners:

1. **Model Status Badge:** Shows `Ready` on CPU/CUDA, indicating the Full AV checkpoint is loaded with its cached validation threshold ($\theta = 0.2950$).
2. **Local Privacy Guarantee:** Emphasize that video and audio processing occur completely on the local machine in a transient directory; no video is retained or uploaded to external clouds.

Step 4: Run Inference and Explain the Output

3. Select a short test clip (under 8 seconds, MP4/WAV).
4. Click **Run Analysis**.
5. Point out the forensic breakdowns:

- **Fused Prediction (p_f):** Overall deepfake probability compared against the validation threshold θ.
- **Modality Probabilities (p_v, p_a):** Independent predictions from the visual head and audio head.
- **Reliability Weights (w_v, w_a):** Show that $w_v + w_a = 1.0$. Point out how the gate assigns dynamic weights based on face quality, audio SNR, and audio-visual synchronization cross-correlation.
- **Quality Indicators:** Display blur variance, audio clipping, and synchronization lag.

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3. EXAMINER Q&A DEFENSE MATRIX (TOP 10 QUESTIONS)

Be fully prepared to answer these specific technical and methodological questions from the examiners:

Q1: "In your validation results, Equal-Weight Fixed Fusion achieved 0.9426 AUROC, while Reliability-Aware Fusion achieved 0.7957. Why did the gate underperform fixed fusion?"

> ***Examiner-Ready Defense:***
>
> *"This is an important, transparent empirical finding. There are two clear scientific reasons:*
>
> *1. **Training Regime:** Our baseline models were trained for only one development epoch. The reliability estimator introduces additional parameters—a 2-layer MLP and a soft-target KL-divergence loss. While the simple fixed-fusion classifier can quickly fit linear combinations of encoder features, the reliability gate requires multi-epoch training to stabilize its weight trajectory.*
>
> *2. **Clean vs. Degraded Data:** In clean, in-domain validation media, both visual and audio signals are pristine and uncorrupted. Under such homogeneous conditions, an equal 50–50 weighting is already near-optimal. The intended advantage of reliability gating is designed for *asymmetric degradation*—such as blurred video, noisy audio, or missing channels. We maintain scientific integrity by reporting these initial validation numbers faithfully rather than disguising them."*

Q2: "Why do you treat audio-visual synchronization as an auxiliary feature rather than a direct classification rule?"

Examiner-Ready Defense:

"Many early works (e.g., Chugh et al., MDS) made the erroneous assumption that desynchronization equals deepfake manipulation. In real-world engineering, this fails in both directions:

*- **False Positives:** Authentic broadcast videos frequently exhibit non-zero temporal offsets due to transmission delays, audio encoding jitter, or professional dubbing.*

*- **False Negatives:** Modern generative tools, such as Wav2Lip, generate synthetic lip movements that are mathematically well-synchronized with the audio track.*

*Therefore, we feed synchronization features (peak cross-correlation, lag dispersion) into our reliability estimator and classifier as *one auxiliary cue among several*, preventing the detector from making naive synchronization-based misclassifications."*

Q3: "Why did the external transfer evaluation on the German MLAAD-tiny slice yield an AUROC of 0.3777?"

Examiner-Ready Defense:

"The MLAAD-tiny evaluation was an intentional, zero-shot external transfer stress test, and the AUROC of 0.3777 illustrates two critical domain-shift phenomena:

*1. **Label Semantic Incompatibility:** In FakeAVCeleb, the model is trained against the clip-level `clip_fake` label, where visual manipulations often occur alongside authentic celebrity audio. In contrast, MLAAD-tiny is an audio anti-spoofing dataset evaluating synthetic voice engines.*

*2. **Strict Evaluation Protocol:** We deliberately kept our decision threshold frozen at $\theta = 0.2950$ (derived from FakeAVCeleb validation) without retraining or fine-tuning on the German target data. When combined with the acoustic phoneme shift between English and German, authentic speech triggered high spoof likelihood. This confirms our core hypothesis that audio-only detectors are fragile to cross-linguistic and cross-task domain shifts."*

Q4: "How did you ensure that there is zero data leakage between your training, validation, and test splits?"

Examiner-Ready Defense:

"We implemented strict source-identity-disjoint partitioning. In FakeAVCeleb, multiple synthetic clips are generated from the same source celebrity identity. If clips from the same identity appear in both train and test splits, the neural network memorizes facial morphology and voice timbre rather than learning generalizable forensic artifacts.

We partitioned the 21,544 clips strictly by celebrity identity: 350 identities in Train (15,018 clips), 50 in Validation (2,136 clips), and 100 in Test (4,390 clips). We verified zero identity overlap and verified zero duplicate file hashes via automated unit tests (`tests/test_split_exclusivity.py` and `tests/test_duplicate_hash.py`). Furthermore, the test split has been kept locked and un-evaluated."

Q5: "Explain the detached soft-target loss $\mathcal{L}_{rel}$ and why you do not just backpropagate directly through the fused classifier."

Examiner-Ready Defense:

"If we optimize modality weights (w_v, w_a) solely via gradient descent through the final classification loss $\mathcal{L}_{BCE}(p_f, y)$, the network suffers from modality competition: one dominant modality (e.g., audio in clean speech) suppresses gradients to the other, causing the gate to permanently collapse into predicting extreme binary weights (e.g., $w_a=1, w_v=0$).

To prevent gate collapse, we formulate an explicit soft-target guide:

$$w_v^* = \frac{\exp(-\mathcal{L}_v / \tau_r)}{\exp(-\mathcal{L}_v / \tau_r) + \exp(-\mathcal{L}_a / \tau_r)}$$

*We *detach* w^* from the computation graph so it acts as an external pseudo-label representing instantaneous unimodal branch performance, and train the reliability MLP using Kullback-Leibler divergence $D_{KL}(w^* \parallel [w_v, w_a])$. This guides the gate toward the modality currently exhibiting lower error without destabilizing the encoders."*

Q6: "Why did you use OpenCV Haar cascades for face extraction instead of a heavier detector like RetinaFace or MediaPipe?"

Examiner-Ready Defense:

"This was a deliberate engineering decision guided by edge-deployment constraints and offline reproducibility. OpenCV Haar detection requires zero external weight downloads, has minimal CPU footprint, and processes 16 frames in under 80 milliseconds. While our Project Brief proposes RetinaFace and landmark alignment for future enhancement, implementing the Haar baseline first allowed us to establish a fully deterministic, self-contained pipeline. We explicitly disclose this limitation in Section 3 and Section 7 of our report."

Q7: "How does your architecture handle missing modalities (e.g., video with muted audio, or audio with occluded video)?"

Examiner-Ready Defense:

"We incorporate explicit missingness indicators $[m_v, m_a]$ directly into the quality feature vectors q_v and q_a. When audio is absent or muted, $m_a=1$, causing the reliability MLP to shift gate weights toward $w_v \to 1.0$ and $w_a \to 0.0$. In the fused classifier, missing

representations are masked to neutral zero vectors. This ensures that the system degrades gracefully into a unimodal visual detector rather than injecting uninformative noise or producing runtime exceptions."

Q8: "What is your validation threshold selection strategy?"

Examiner-Ready Defense:

"We select the decision threshold θ^* on the validation set using Youden's J statistic ($J = \text{Sensitivity} + \text{Specificity} - 1$), which balances true positive rate against false positive rate under class imbalance. Because FakeAVCeleb has an extreme class imbalance (~97.7% fake), evaluating at a default $\theta=0.5$ produces skewed accuracy. We fit the threshold on validation and lock it before inference, strictly forbidding threshold re-tuning on evaluation benchmarks."

Q9: "What are the computational specifications and latency of your model?"

Examiner-Ready Defense:

"The visual encoder (MobileNetV3-Small) has approximately 1.5 million parameters, and the audio DS-CNN has under 800,000 parameters. The total model footprint is under 3 million parameters (< 15 MB checkpoint size). On standard CPU hardware without GPU acceleration, the complete end-to-end inference—including audio decoding, face detection, Mel-spectrogram computation, and fused forward pass—executes in approximately 350 to 500 milliseconds for a 2-second clip, comfortably satisfying near-real-time streaming constraints."

Q10: "What steps were taken to guarantee academic integrity and low Turnitin plagiarism?"

Examiner-Ready Defense:

"Our final report was drafted from first principles without relying on generative text recycling or copying from published IEEE abstracts. We structured our literature review into an analytical synthesis matrix, formatted all mathematical formulas in original LaTeX notation, adhered to standard IEEE numerical citations, and verified that our Turnitin similarity index remains well below the institutional threshold of 10% with bibliography filtering enabled."